MEC6602E Transonic Aerodynamics : 2D Euler basic scheme

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Vector form

$$\Omega \frac{\partial \vec{W}}{\partial t} + \sum_{m=1}^{N_F} (\vec{F}_c)_m S_m = 0$$

$$\vec{W} = \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ \cancel{\rho w} \\ \rho E \end{pmatrix}_{\text{3D}} \qquad \vec{F}_c = \begin{pmatrix} \rho V_c \\ \rho u V_c + n_x p \\ \rho v V_c + n_y p \\ \cancel{\rho w V_c + n_z p} \\ \rho E V_c + p V_c \end{pmatrix}_{\text{3D}}$$

Contravariant velocity: $V_c = \vec{V} \cdot \vec{n}$

References

Algorithms shown in this presentation are taken from:

- Blazek, J. (2015). *Computational Fluid Dynamics: Principles and Applications*. Butterworth-Heinemann.

This is a complete reference for CFD methods.

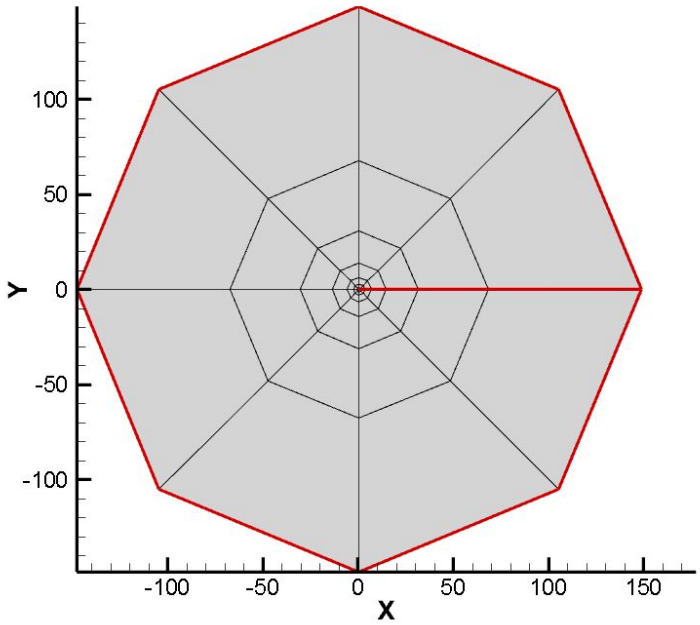
Structured plot3D file format

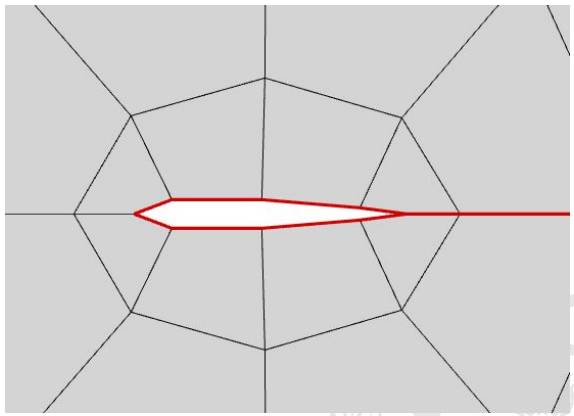
Meshes are provided on Moodle in plot3D format:

- An index for the number of blocks m;
- m sets of 3 indexes n_i n_j;
- A list of coordinates in the order:
  - All x of block 1 (order i, j, k);
  - All y of block 1 (order i, j, k);
  - All x of block 2 (order i, j, k);
  - etc.

Meshes are provided from 9×9 to 2049×2049.

Add 2 layers of ghost cells for the boundary conditions implementation





Steps of a CFD solver

1. Read the mesh
2. Compute the geometrical quantities (cell volume, face surface, face normal, etc.)
3. Set the initial and boundary conditions
4. Iterate over time (until convergence or desired time):
   a. Loop on faces to compute the fluxes;
   b. Compute the residual $\vec{R}$ of each cell;
   c. Advance $\vec{W}$ in time (explicitly or implicitly);
   d. Update the boundary conditions;
   e. Compute the forces and moments (integrate the pressure at the surface);
   f. Monitor convergence
5. Post process (write the field solution, the solid wall solution, the force and moment coefficients).

This slide's topics: 4(b)–4(d)

Convective fluxes discretization

We know from the wave equation that the numerical scheme must be either upwinded or added of a numerical dissipation.
Several schemes can be used to discretize the convective scheme.

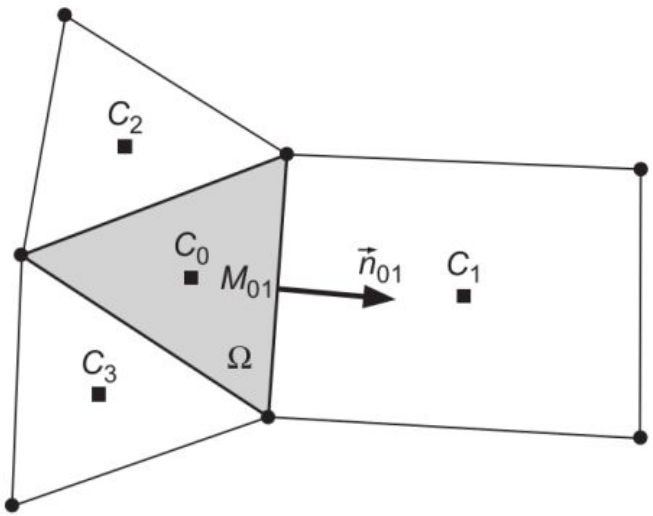
> **Important:** To avoid issue from truncation error, it is better to compute the flux at the face and add it to the residual of both adjacent cell. If the flux is computed in a loop on the cells it will be computed twice.

When adding the flux to the right and left state, you must take account of the normal direction. You must invert the flux for one cell. What is leaving the left cell must enter the right cell.

Centered fluxes discretization

$$\#1 \quad (\vec{F}_c \Delta S)_{01} \approx \frac{1}{2}\left[\vec{F}_c(\vec{W}_0, \vec{n}_{01}) + \vec{F}_c(\vec{W}_1, \vec{n}_{01})\right]\Delta S_{01}$$

$$\#2 \quad (\vec{F}_c \Delta S)_{01} \approx \vec{F}_c(\vec{W}_{01}, \vec{n}_{01})\,\Delta S_{01}, \qquad \vec{W}_{01} = \frac{1}{2}(\vec{W}_0 + \vec{W}_1)$$



The two methods are *not* equivalent. First method is better for interior faces. However, second method is to be used at physical boundaries (except the connectivity, which must behave like interior faces).

Central Scheme with Scalar Dissipation

Convective flux:

$$(\vec{F}_c\,\Delta S)_{I+1/2,J,K} \approx \vec{F}_c(\vec{W}_{I+1/2,J,K})\,\Delta S_{I+1/2,J,K} - \vec{D}_{I+1/2,J,K}$$

Centered scheme:

$$\vec{F}_c(\vec{W}_{I+1/2,J,K}) = \vec{F}_c(\tfrac{1}{2}(\vec{W}_{I,J,K} + \vec{W}_{I+1,J,K})) \quad \text{(see previous slide)}$$

Artificial Dissipation:

$$\vec{D}_{I+1/2} = \hat{\Lambda}^S_{I+1/2}\Big[\underbrace{\epsilon^{(2)}_{I+1/2}(\vec{W}_{I+1} - \vec{W}_I)}_{\text{2nd order term}} - \underbrace{\epsilon^{(4)}_{I+1/2}(\vec{W}_{I+2} - 3\vec{W}_{I+1} + 3\vec{W}_I - \vec{W}_{I-1})}_{\text{4th order term}}\Big]$$

Notice the switch in sign between second and fourth-order dissipation.
The equation is written for I direction. It must be applied in the J direction as well.

Central Scheme with Scalar Dissipation

The constant in front of the dissipation terms are scaled so that:

- The fourth-order term is switched off across shocks;
- The second-order term is switched off in smooth region of the flow field.

$$\epsilon^{(2)}_{I+1/2} = k^{(2)} \max(\Upsilon_I, \Upsilon_{I+1})$$

$$\epsilon^{(4)}_{I+1/2} = \max\left[0, (k^{(4)} - \epsilon^{(2)}_{I+1/2})\right]$$

$$\Upsilon_I = \frac{|p_{I+1} - 2p_I + p_{I-1}|}{p_{I+1} + 2p_I + p_{I-1}}$$

$$k^{(2)} = 1/2, \qquad 1/128 \leq k^{(4)} \leq 1/64$$

Central Scheme with Scalar Dissipation

The dissipation is scaled by a SCALAR value, the spectral radius. The spectral radius is the largest eigenvalue of the jacobian matrix (the largest information convection speed at a given point).

$$\hat{\Lambda}_c = (|V| + c)\Delta S$$

$$(\hat{\Lambda}_c^I)_{I+1/2} = \tfrac{1}{2}\left[(\hat{\Lambda}_c^I)_I + (\hat{\Lambda}_c^I)_{I+1}\right], \qquad \hat{\Lambda}^S_{I+1/2} = (\hat{\Lambda}_c^I)_{I+1/2} + (\hat{\Lambda}_c^J)_{I+1/2}$$

$$\hat{\Lambda}_c^I = (|\vec{V}\cdot\vec{n}^I| + c)\Delta S^I, \qquad \hat{\Lambda}_c^J = (|\vec{V}\cdot\vec{n}^J| + c)\Delta S^J$$

V is the contravariant velocity (a scalar).
Note the area of the surface integral is lumped inside the spectral radius.

How to compute the Cell-centered spectral radii

$$\hat{\Lambda}_c^I = (|\vec{v} \cdot \vec{n}^I| + c)\Delta S^I, \qquad \hat{\Lambda}_c^J = (|\vec{v} \cdot \vec{n}^J| + c)\Delta S^J$$

The surface normal and the area are quantities related to the faces of the mesh.
To compute them at the cell-center, average the values from the two faces bounding the cell in the I, J and K direction.
The velocity and speed of sound are known at the cell center.

Central Scheme with Matrix Dissipation

A less diffusive scheme can be obtained by using the Jacobian matrix to scale the dissipation instead of the spectral radius.

$$\vec{D}_{I+1/2} = |\bar{A}_c|_{I+1/2} \left[\epsilon^{(2)}_{I+1/2} (\vec{W}_{I+1} - \vec{W}_I) - \epsilon^{(4)}_{I+1/2} (\vec{W}_{I+2} - 3\vec{W}_{I+1} + 3\vec{W}_I - \vec{W}_{I-1}) \right]$$

$$|\bar{A}_c| = \bar{T} \, |\bar{\Lambda}_c \, \Delta S| \, \bar{T}^{-1}$$

- $\bar{T}$ – The matrix of right eigenvector of the Jacobian matrix
- $\bar{T}^{-1}$ – The matrix of left eigenvector of the Jacobian matrix
- $\bar{\Lambda}_c$ – The diagonal matrix of the Jacobian matrix eigenvalues

The matrix $|\bar{A}_c|_{I+1/2}$ can be efficiently computed by the method provided in: Swanson, R. C., & Turkel, E. (1992). On central-difference and upwind schemes. *Journal of Computational Physics*, 101(2), 292–306.

Boundary Conditions

We implement the boundary conditions by setting the value in the ghost cells so to compute the correct flux at the boundary face. This is why we take the average of the cell value at the face, then compute the flux. Instead of the average of the flux.

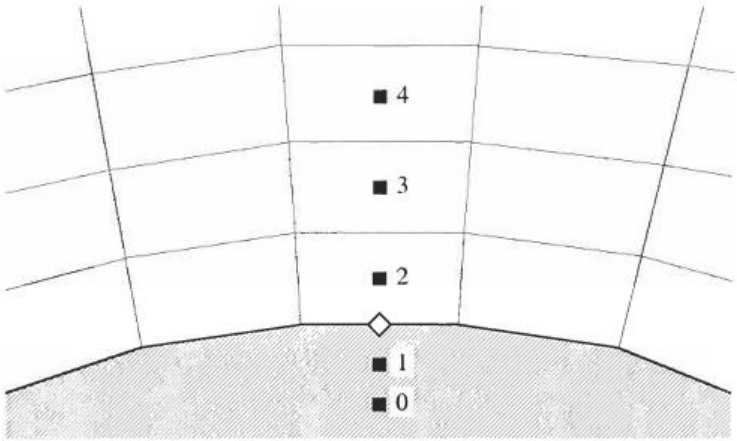
Beware of the direction of the normal vectors. Direction are often taken towards the outside of the domain in textbooks.

No-Slip Wall Boundary Condition

We want $\vec{V} \cdot \vec{n} = 0$ at the wall

$$\vec{v_1} = \vec{v_2} - 2V_2\vec{n}$$
$$V_2 = u_2 n_x + v_2 n_y + w_2 n_z$$

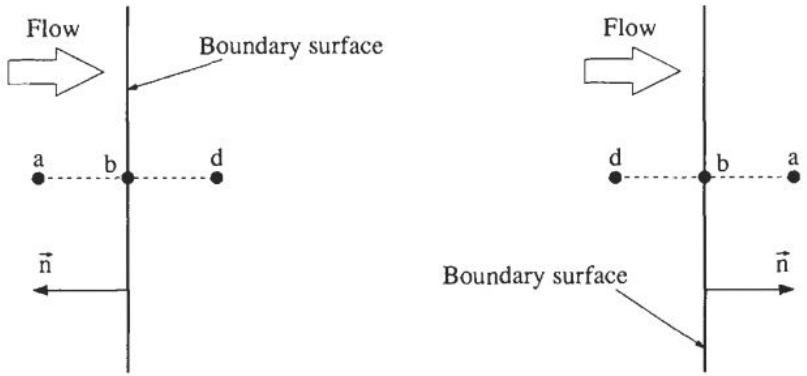
$$P_1 = P_2$$
$$P_0 = P_2$$
$$\rho_1 = \rho_2$$
$$\rho_0 = \rho_2$$



Farfield

Note: the b node is the boundary, we use an extrapolation to set the value in the ghost cell center

Supersonic inflow: $\vec{W}_b = \vec{W}_a$
Supersonic outflow: $\vec{W}_b = \vec{W}_d$



Subsonic inflow:

$$p_b = \frac{1}{2}\{p_a + p_d - \rho_0 c_0[n_x(u_a - u_d) + n_y(v_a - v_d) + n_z(w_a - w_d)]\}$$
$$\rho_b = \rho_a + (p_b - p_a)/c_0^2$$
$$u_b = u_a - n_x(p_a - p_b)/(\rho_0 c_0), \quad v_b = v_a - n_y(p_a - p_b)/(\rho_0 c_0)$$
$$w_b = w_a - n_z(p_a - p_b)/(\rho_0 c_0)$$

Farfield (continued)

Subsonic outflow:

$$
\begin{aligned}
p_b &= p_a \\
\rho_b &= \rho_d + (p_b - p_d)/c_0^2 \\
u_b &= u_d + n_x(p_d - p_b)/(\rho_0 c_0), \quad v_b = v_d + n_y(p_d - p_b)/(\rho_0 c_0) \\
w_b &= w_d + n_z(p_d - p_b)/(\rho_0 c_0)
\end{aligned}
$$

ρ_0 and c_0 are taken as the state at the interior point. Values in a are the values from the freestream. Values are then extrapolated from the interior and the BC to the ghost cell centers.

Explicit time integration

The simplest time integration scheme is the explicit Euler one:

$$\Omega_i \frac{\partial \vec{W}_i}{\partial t} + R_i = 0$$

$$\Omega_i \frac{\vec{W}_i^{n+1} - \vec{W}_i^n}{\Delta t_i} + R_i^n = \Omega_i \frac{\Delta \vec{W}_i}{\Delta t_i} + R_i^n = 0$$

$$\Delta \vec{W}_i = -R_i^n \frac{\Delta t_i}{\Omega_i}$$

$$\Delta t_I = \sigma \frac{\Omega_I}{(\hat{\Lambda}_c^I + \hat{\Lambda}_c^J + \hat{\Lambda}_c^K)_I}$$

This scheme is limited to CFL (σ) lower than 1

$$\hat{\Lambda}_c^I = (|\vec{v} \cdot \vec{n}^I| + c)\Delta S^I$$
$$\hat{\Lambda}_c^J = (|\vec{v} \cdot \vec{n}^J| + c)\Delta S^J$$
$$\hat{\Lambda}_c^K = (|\vec{v} \cdot \vec{n}^K| + c)\Delta S^K$$

Explicit Multistage scheme

$$\vec{W}_I^{(0)} = \vec{W}_I^n$$

$$\vec{W}_I^{(1)} = \vec{W}_I^{(0)} - \alpha_1 \frac{\Delta t_I}{\Omega_I} R_I^{(0)}$$

$$\vec{W}_I^{(2)} = \vec{W}_I^{(0)} - \alpha_2 \frac{\Delta t_I}{\Omega_I} R_I^{(1)}$$

$$\vdots$$

$$\vec{W}_I^{n+1} = \vec{W}_I^{(m)} = \vec{W}_I^{(0)} - \alpha_m \frac{\Delta t_I}{\Omega_I} R_I^{(m-1)}$$

	1st-order scheme			2nd-order scheme		
stages	3	4	5	3	4	5
σ (CFL)	1.5	2.0	2.5	0.69	0.92	1.15
α_1	0.1481	0.0833	0.0533	0.1918	0.1084	0.0695
α_2	0.4000	0.2069	0.1263	0.4929	0.2602	0.1602
α_3	1.0000	0.4265	0.2375	1.0000	0.5052	0.2898
α_4		1.0000	0.4414		1.0000	0.5060
α_5			1.0000			1.0000

Hybrid Multistage Schemes

We split the residual vector: $R_I = (R_c)_I - (R_d)_I$.

$$(R_c)_I = \sum_{k=1}^{N_F} \left[\vec{F}_c(\vec{W}_{av})\Delta S\right]_k - \underbrace{(\vec{Q}\,\Omega)_I}_{\text{source terms}}, \qquad (R_d)_I = \sum_{k=1}^{N_F} \underbrace{\left[\vec{F}_v\,\Delta S + \vec{D}\right]_k}_{\text{viscous fluxes (N.-S.)}}$$

	central, σ=3.6		upwind, σ=2.0	
stage	α	β	α	β
1	0.2500	1.00	0.2742	1.00
2	0.1667	0.00	0.2067	0.00
3	0.3750	0.56	0.5020	0.56
4	0.5000	0.00	0.5142	0.00
5	1.0000	0.44	1.0000	0.44

$$\vec{W}_I^{(0)} = \vec{W}_I^n$$

$$\vec{W}_I^{(1)} = \vec{W}_I^{(0)} - \alpha_1 \frac{\Delta t_I}{\Omega_I}\left[R_c^{(0)} - R_d^{(0)}\right]_I$$

$$\vec{W}_I^{(2)} = \vec{W}_I^{(0)} - \alpha_2 \frac{\Delta t_I}{\Omega_I}\left[R_c^{(1)} - R_d^{(0)}\right]_I$$

$$\vec{W}_I^{(3)} = \vec{W}_I^{(0)} - \alpha_3 \frac{\Delta t_I}{\Omega_I}\left[R_c^{(2)} - R_d^{(2,0)}\right]_I$$

$$\vec{W}_I^{(4)} = \vec{W}_I^{(0)} - \alpha_4 \frac{\Delta t_I}{\Omega_I}\left[R_c^{(3)} - R_d^{(2,0)}\right]_I$$

$$\vec{W}_I^{n+1} = \vec{W}_I^{(0)} - \alpha_5 \frac{\Delta t_I}{\Omega_I}\left[R_c^{(4)} - R_d^{(4,2)}\right]_I$$

$$R_d^{(2,0)} = \beta_3 R_d^{(2)} + (1-\beta_3) R_d^{(0)}$$

$$R_d^{(4,2)} = \beta_5 R_d^{(4)} + (1-\beta_5) R_d^{(2,0)}$$

Homework 2 – Next Steps

1. Implement the convective flux scheme;
2. Implement the solution update by the explicit Euler scheme;
3. Implement the supersonic boundary condition (you can do a sanity check by setting all the BC as farfield, with solution initialized as free stream – nothing should happen);
4. Implement Slip Wall and connectivity BC;
5. Implement the multistage scheme;
6. Implement the Subsonic boundary condition.

This would be a functioning code for the Homework 2. Then we can improve on this.

Next Week

Numerical Scheme

- Second Order Roe Scheme

Acceleration techniques:

- Local time stepping;
- Residual Smoothing;
- Multigrid method;
- Implicit Scheme;
- Enthalpy damping.